

Coherent Kinematics in Modal Triplet Theory

Chart Persistence, Encoding Trajectories, and Physical Bridges

Peter Nero

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Version 2

Abstract

We formulate coherent kinematics as an encoding-level construction and separate it from physical causal motion. An admissible chart is a typed reduction from an upper state domain to an effective description, with transition maps on controlled overlaps. Position is a declared localization functional of chart support. An encoding trajectory is generated by an actual upper path that remains representable through a compatible chain of charts; equivalent chart descriptions define the same encoding worldline. Regularity of this worldline requires regular upper evolution and regular chart and localization maps—contractivity alone is insufficient. Physical timelike and null classification, finite propagation, and locality come only after a selected local hyperbolic realization and its principal symbol are supplied. Chart ordering alone is not a causal order. Effective mergers are noninjectivity of an autonomous reduced map, while a split requires additional multivalued, stochastic, or hybrid continuation data. At a horizon, failure of an exterior decoder or chart continuation does not imply termination of the underlying physical worldline. These distinctions retain chart persistence as a useful MTT kinematic language without deriving relativity, exclusion principles, measurement outcomes, or irreversibility from projection alone.

Version 2 Revision Note

Supersedes The first edition of *Coherent Kinematics in Modal Triplet Theory*.

Reason Chart persistence, partial order, noninjective projection, and decoder failure were overinterpreted as physical motion, causal order, mergers, splits, and worldline termination.

Resolution Version 2 defines chart-relative position and encoding worldlines, states their regularity gates, and requires a separate hyperbolic bridge for physical causality and finite propagation.

Retained result Compatible chart chains still define a useful encoding-level kinematics.

Remaining boundary Physical trajectories, horizons, arrows of time, and the relation of shared phase to time require the selected physical law.

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1 Three levels of kinematic statement

We distinguish:

Upper evolution. A path or flow in the declared MTT control/state space.

Encoding kinematics. Localization and persistence after applying compatible partial descriptions.

Physical kinematics. Curves, support propagation, and causal classes in a selected space-time evolution law.

Encoding kinematics can represent physical motion after a bridge theorem. It does not itself provide a metric, null cone, velocity, force law, or physical time. The stabilization parameter τ , a physical time t , and an RG scale μ remain separately typed.

2 Admissible charts and transitions

Let X be an upper state space. An admissible chart is a triple

$$\mathfrak{C}_\alpha = (A_\alpha, Y_\alpha, r_\alpha), \quad r_\alpha : A_\alpha \rightarrow Y_\alpha,$$

where $A_\alpha \subseteq X$ is a domain on which every declared operator, projection, stability, and interpretation margin is controlled. The map r_α is a reduction or encoding map; it need not be injective, linear, or a coordinate chart on X .

On $A_{\alpha\beta} = A_\alpha \cap A_\beta$, an exact transition map is a map $h_{\beta\alpha}$ on the attained overlap image satisfying

$$r_\beta(x) = h_{\beta\alpha}(r_\alpha(x)), \quad x \in A_{\alpha\beta}.$$

Existence requires r_β to be constant on the r_α fibers inside the overlap, by the Projection–Admissibility descent theorem. Approximate compatibility must state a metric, an error bound, and an accumulation rule.

Assumption 2.1 (Compatible chart family). On every nonempty overlap, the required transition maps exist, are continuous in the chosen topologies, obey $h_{\alpha\alpha} = \text{id}$, and satisfy the cocycle identity

$$h_{\gamma\beta}h_{\beta\alpha} = h_{\gamma\alpha}$$

where all terms are defined.

This assumption constructs an effective atlas of descriptions. It does not turn the upper state space into physical spacetime.

3 Coherent carriers, support, and position

A coherent carrier $\mathcal{B}$ is a declared upper subset or labeled family controlled by the corrected Foundation and Fixed Points hypotheses. Depending on the application it may be an invariant basin, a localized solution family, or a neighborhood of a selected equilibrium. These cases are not identified without a theorem.

Its attained support in chart α is

$$\text{Supp}_\alpha(\mathcal{B}) := r_\alpha(\mathcal{B} \cap A_\alpha).$$

Because r_α may be noninjective, equal supports do not prove equality of upper carriers. Conversely, different chart supports may describe the same carrier through a transition map. No exclusion principle follows from support coincidence alone.

Definition 3.1 (Chart-relative position). Let $\mathcal{S}_\alpha$ be a declared class of localized support sets in Y_α . A position diagnostic is a map

$$\ell_\alpha : \mathcal{S}_\alpha \rightarrow Z_\alpha$$

with specified covariance and regularity under chart transitions. The chart-relative position of $\mathcal{B}$ is $\ell_\alpha(\text{Supp}_\alpha \mathcal{B})$ when that support belongs to $\mathcal{S}_\alpha$.

Centroids require further geometry. On a Riemannian base they should be intrinsic Karcher means inside a convex normal ball, as in FP IV, rather than unqualified coordinate averages.

4 Encoding trajectories and worldline equivalence

Let J be an interval of an explicitly named parameter s , and let $x : J \rightarrow X$ be an upper path. This path may be a stabilization trajectory, a physical solution lifted to the upper space, or another declared family.

Definition 4.1 (Chart-persistent encoding trajectory). The path x has a chart-persistent encoding on J if there is an interval cover $\{J_i\}$ and chart labels α_i such that $x(J_i) \subseteq A_{\alpha_i}$ and

$$z_i(s) = r_{\alpha_i}(x(s))$$

satisfies

$$z_j(s) = h_{\alpha_j\alpha_i}(z_i(s))$$

on every time and chart overlap.

Definition 4.2 (Encoding worldline equivalence). Two chart-persistent families represent the same encoding worldline if their curves agree through the declared transition maps on a common refinement of their interval covers. A monotone reparameterization may also be quotiented out when the application declares parameter-free curve equivalence.

Theorem 4.3 (Gluing of encoding trajectories). *Under Assumption 2.1, a chart-persistent family defines a unique curve in the quotient effective atlas, up to the stated worldline equivalence. Conversely, a curve in that atlas lifts locally to compatible chart curves, but not necessarily to a unique upper path.*

Proof. The transition and cocycle equations identify all chart representatives on double and triple overlaps, producing a quotient curve. Local representatives give the converse. Noninjectivity of the chart maps prevents unique upper recovery in general. $\square$

5 Regularity requires more than contraction

Theorem 5.1 (Regular encoding trajectory). *Suppose $x : J \rightarrow X$ is locally Lipschitz, each active r_α is Lipschitz on the attained path, each transition map is Lipschitz on the attained overlap, and each localization diagnostic ℓ_α is Lipschitz with respect to a declared support metric. Then every chart and position representative of the encoding worldline is locally Lipschitz, with constants bounded by the product of the corresponding bounds.*

Proof. Compose the Lipschitz upper path with the chart and localization maps. On overlaps, compose once more with the Lipschitz transition map. $\square$

A contraction estimate for a time-step map controls separation of iterates in the chosen state norm. By itself it proves neither continuity of the parameter path nor continuity of a chart, support map, or localization functional. Those regularity assumptions cannot be omitted.

6 Bridge to physical spacetime motion

Adopt the canonical physical realization only in this section: a globally hyperbolic Lorentzian base (Y_4, g) and a selected local physical equation with well-posed Cauchy evolution. Its gauge-fixed principal symbol determines the characteristic cone. The corrected Signature Stability paper states the conditions under which coherent compression preserves that supplied cone.

Definition 6.1 (Physical causal class). A differentiable physical curve $\gamma : I \rightarrow Y_4$ is timelike, null, or spacelike according to the sign of $g(\dot{\gamma}, \dot{\gamma})$. This definition belongs to the selected Lorentzian realization, not to chart overlap alone.

Theorem 6.2 (Conditional finite propagation). *Suppose the selected physical equation has a finite-domain-of-dependence theorem with causal cone g , and let a localized solution have initial support K on a Cauchy surface. Then its later support is contained in $J^+(K) \cup J^-(K)$ as appropriate to the evolution interval. Coherent compression preserves this conclusion when the principal-symbol descent and locality hypotheses hold.*

This is the available no-superluminal-support statement. It is conditional on the physical PDE and concerns support propagation. It is not proved by admissibility, contractivity, or a partial order of charts.

A localization diagnostic may define a representative physical worldline only when existence, uniqueness, and regularity of that localization are proved. Wave packets and extended supports need not have a unique point trajectory.

7 Chart order is not causal order

One may partially order charts by domain inclusion, refinement, or allowed continuation. Such an order records descriptive availability. A physical causal order records reachability by causal curves or support propagation in a Lorentzian/hyperbolic realization. Neither order determines the other without a monotone bridge theorem.

Likewise, an oriented stabilization flow can provide an ordering in τ but not a physical arrow of time. A temporal arrow requires a physical evolution plus an asymmetric law or boundary condition, such as a Lyapunov or entropy theorem in its proper domain.

A compact shared $U(1)$ circle may synchronize phase or holonomy across modal lanes. Its angular coordinate is not physical time. Relating it to temporal ordering requires a declared lift $\mathbb{R} \rightarrow S^1$, a selected noncompact evolution parameter, and a theorem showing how the lifted order enters the hyperbolic physical propagator.

8 Effective mergers, splits, and identity

Suppose autonomous descent gives a reduced map $F_s : Y_0 \rightarrow Y_s$.

Definition 8.1 (Effective merger). An effective merger occurs when $y_1 \neq y_2$ but $F_s(y_1) = F_s(y_2)$. The final effective state does not uniquely decode the prior effective label.

This is a statement about the reduced map. It does not prove physical coalescence, attractive interaction, destruction of either upper state, or entropy production.

A deterministic single-valued map sends one state to one output and cannot by itself encode a split into alternative outcomes. A split description requires a refined state space, a multivalued relation, a stochastic kernel with a measure, or a hybrid branch/reset law. Branch probabilities do not follow from the number of chart continuations.

Identity across charts is therefore a declared equivalence supported by transition data and diagnostics. If the diagnostic family is not separating, indistinguishability does not prove upper identity.

9 Horizons and exterior descriptions

Let r_{ext} restrict a global physical solution to an exterior observable domain. It may be noninjective. A section of r_{ext} chooses one constraint-compatible global extension; an exact decoder would recover the actual global solution and generally requires injectivity.

An event horizon in a regular globally hyperbolic spacetime does not generally terminate an infalling timelike worldline at horizon crossing. What may end is its representation in a chosen exterior chart or the ability of exterior data to decode interior information. Singular endpoints, geodesic incompleteness, and chart boundaries are separate notions.

Thus horizon language in MTT must specify whether the failure is:

1. loss of an exterior chart;
2. loss or ill-conditioning of a representative extension;

3. failure of an exact exterior decoder;
4. breakdown of the physical PDE; or
5. genuine physical geodesic incompleteness.

No global-right-inverse obstruction follows from noninjectivity alone.

10 Relation to the corrected MTT spine

The Foundation supplies typed state spaces, coherent projection, separate evolution parameters, and admissibility margins. Fixed Points I–VI supply conditional fixed points, equilibrium promotion, damping, leakage, intrinsic centroids, and frozen covariance. Projection–Admissibility supplies the descent, section, decoder, and merger types. Signature Stability supplies the physical principal-symbol bridge.

Consequently:

- chart persistence is a valid encoding-level construction;
- physical locality and null/timelike classes come from the selected hyperbolic realization;
- effective merger does not replace the FP IV signed-interaction and dynamical-connection requirements; and
- admissibility exit does not select a successor chart or physical state.

Quantum measurement, photons, QFT propagation, and cosmological horizons may instantiate this framework only after their own physical state, evolution, and observable maps have been constructed. They are not derived here.

11 Scoped coherent-kinematics theorem

Theorem 11.1 (Encoding kinematics with a physical bridge). *Assume a compatible admissible chart family and a regular upper path satisfying Theorem 5.1. Then the path defines a regular encoding worldline up to chart-transition equivalence. If, additionally, the upper path comes from a localized solution of a selected globally hyperbolic physical equation and the coherent principal-symbol/locality descent hypotheses hold, the encoding can represent physical support propagation within the supplied causal cone.*

The first conclusion is encoding kinematics. The second is a conditional physical bridge. Neither conclusion derives the metric, the physical equation, particle identity, a unique centroid, a merger outcome, or a temporal arrow.

12 Conclusion

Chart-relative position and persistence remain useful MTT concepts once their status is kept precise. They describe how an upper carrier is represented through compatible partial encodings. Physical motion requires more: a local hyperbolic equation, a causal principal symbol, finite propagation, and a controlled localization map. With this distinction, horizon, merger, and split language can be used without confusing loss of description with physical termination or selection.

References

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